

1-126
128-191
192-223

A
B
C

camlin

Date : _____

Q Find the net id and host id of the following IP addresses.

a. 74.23.100.203

b. 195.168.10.32

c. 150.40.30.211

d. 111.200.15.35

e. 210.15.190.200

f. 170.90.11.89

a. 74.23.100.203 - belongs to class A

net id : 74

74

host id : 23.100.203

23.100.203

b. 195.168.10.32 - belongs to class B

net id : 195.168.10

195.168.10

host id : 32

32

c. 150.40.30.211 - belongs to class B

net id : 150.40

150.40

host id : 30.211

30.211

d. 111.200.15.35 - belongs to class A

net id : 111

111

host id : 200.15.35

200.15.35

e. 210.15.190.200 - belongs to class C

net id : 210.15.190

host id : 200

f. 170.90.11.89 - belongs to class B

net id : 170.90

host id : 11.89

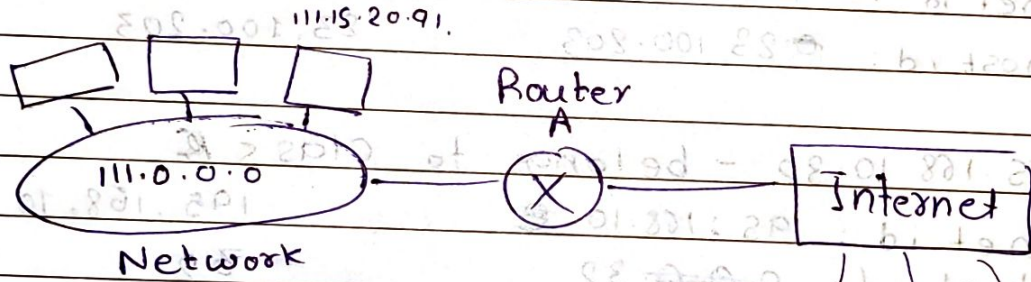
* Network Address:

SIP | DIP

routing is done based on destination IP address. The network address is first found from the DIP. This is called masking (& operation)

0
↓ Masking Bit

0/1 → AND → 0



To get net id we mask out host id bits

	Default Mask
Class A	255.0.0.0
Class B	255.255.0.0
Class C	255.255.255.0

Net id = ip address AND default Mask

first address of the block.
ex. 111.0.0.0, 112.0.0.0

Initial address / first add cannot be assigned to any machine

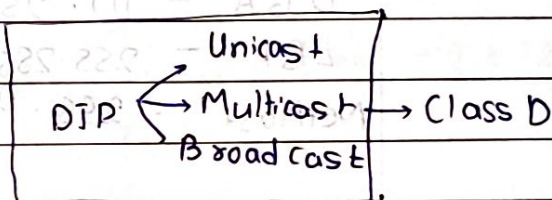
111.0.0.0
111.255.255.255 block

Q Find Network address / Initial address / block address of the following IP address.

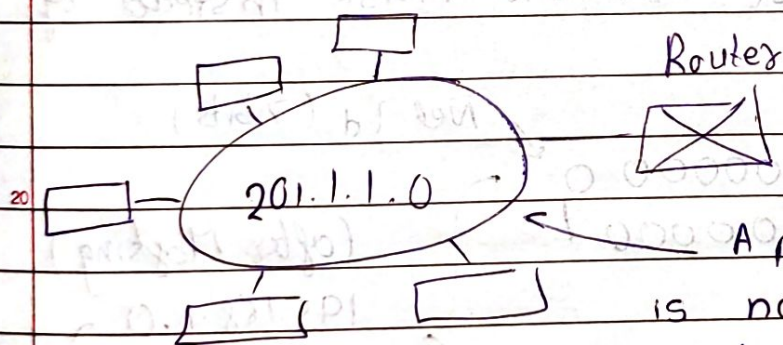
- | | | | |
|----|---------------|---|-------------|
| 1. | 201.16.32.195 | - | 201.16.32.0 |
| 2 | 150.90.200.35 | - | 150.90.0.0 |
| 3 | 90.200.135.45 | - | 90.0.0.0 |
| 4 | 199.168.4.244 | - | 199.168.4.0 |
| 5 | 115.15.20.89 | - | 115.0.0.0 |
| 6 | 140.90.131.50 | - | 140.90.0.0 |

* Broadcast address.

1. Limited broadcast
2. Directed broadcast.



LBA : 255.255.255.255



A packet with DIP 255.255.255.255

is not allowed to leave the network by router.

Hence it is called limited broadcast address (LBA).

(This broadcast happens in own network itself)

- In directed broadcast the source of information is out of the network.

**

The router uses directed broadcast address when packet arrives from out of the network.

SIP. 201.1.1.254

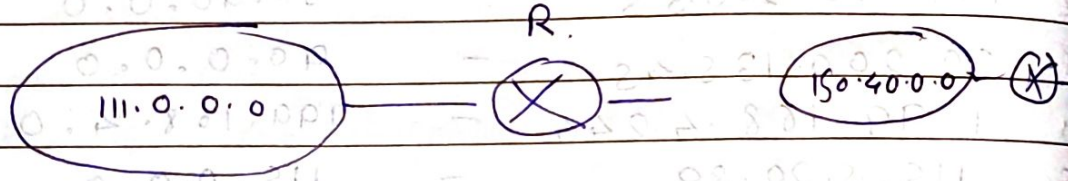
DIP. 201.1.1.255

Ex.

Class A: 111.0.0.0 - 111.255.255.255.

First Address

Last Address.



Network Add - 111.0.0.0

150.40.0.0

DBA - 111.255.255.255

150.40.255.255

LBA - 255.255.255.255

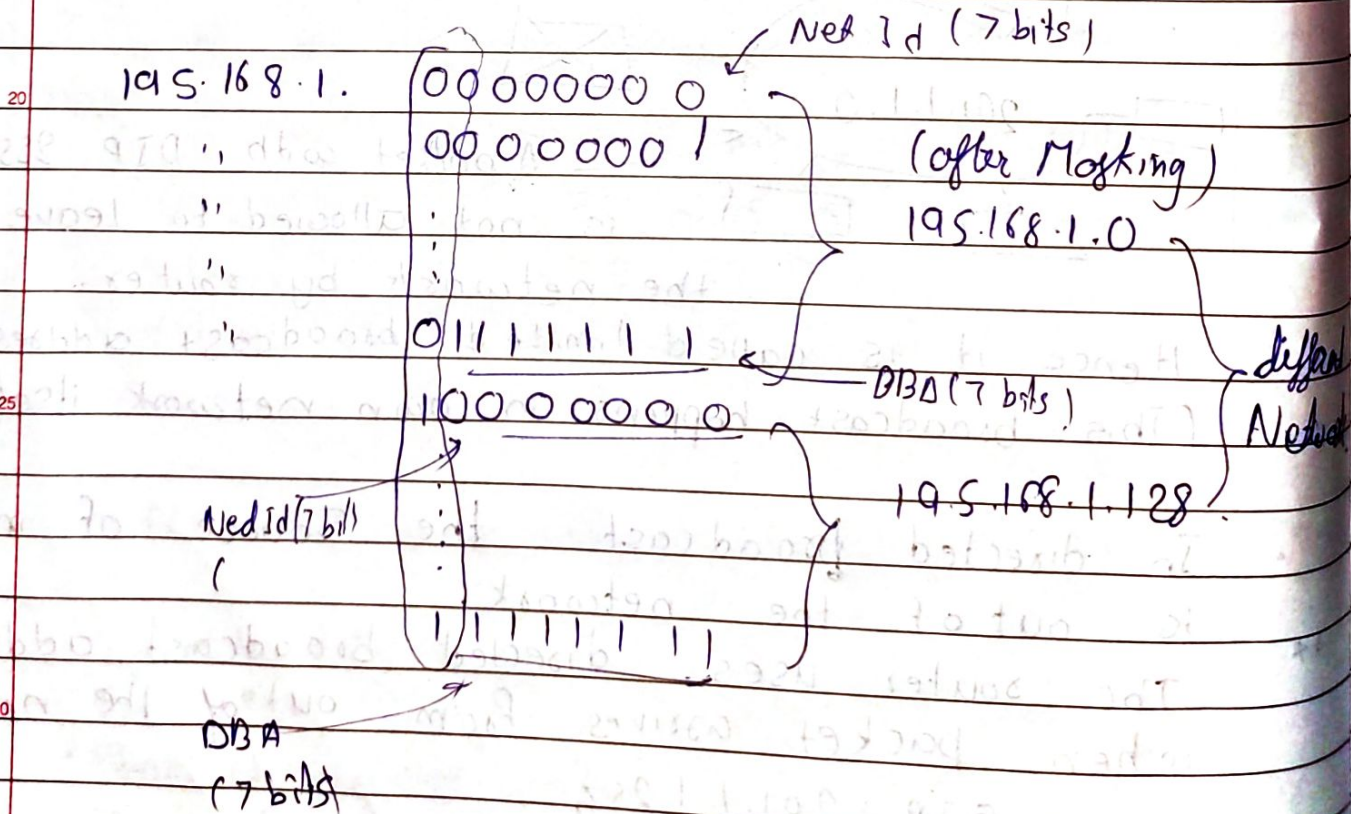
255.255.255.255

NetMask - 255.0.0.0

255.255.0.0

* 15 Subnetting:

For class C use 25 bit mask instead of 24.



11000000
10000000
10000000
10000000

Design 4 Subnets. and find S1, S2, S3, S4 for IP
195.168.1.0

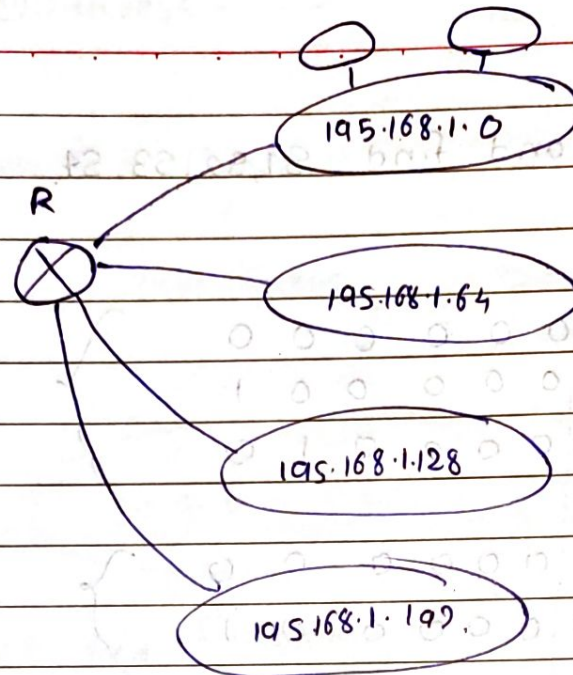
(After Masking)

5	195.168.1.	0 0	0 0	0 0	0 0	0 0	0 0	0 0	195.168.1.0
	195.168.1.	0 0	0 0	0 0	0 0	0 0	0 0	1	
	195.168.1.	0 0	0 0	0 0	0 0	0 0	0 0	1 0	
		0 1	0 0	0 0	0 0	0 0	0 0	0 0	195.168.1.64
10		0 1	0 0	0 0	0 0	0 0	0 0	1	
		1 0	0 0	0 0	0 0	0 0	0 0	0 0	195.168.1.128
		1 0	0 0	0 0	0 0	0 0	0 0	1	
15		1 1	0 0	0 0	0 0	0 0	0 0	0 0	
		1 1	0 0	0 0	0 0	0 0	0 0	1 0	195.168.1.192
		1 1	1 1	1 1	1 1	1 1	1 1	1 1	

NetMask: 255.255.255.192
length of Subnet mask = 26

S₁ = Subnet 1: Net ID: 195.168.1.0 to 195.168.1.63
S₂ = Subnet 2: 195.168.1.64 to 195.168.1.127
S₃ = Subnet 3: 195.168.1.128 to 195.168.1.191
S₄ = Subnet 4: 195.168.1.192 to 195.168.1.255

for Network without Subnet we can use 256-2 address
(where 1 is NetID and bcast is direct broadcast)
but when we use 4 Subnet we can only use
256-8 address. as each Subnet has
its own Sub. netID and DBA



If Source IP and Destination IP belong to same network (Subnet) the packet does not go to router. Instead it is delivered directly in home network.

Q B class IP divide the network in 64 subnets and give the range of first 3 and last 3 subnet IP.

Lets Assume our network block of class B is 150.1.0.0 to 150.1.255.255.

NetMask : 255.255.252.0

S₁ : 150.1.0.0

S₂ :

S₃ :

S₆₂ :

S₆₃ :

S₆₄ :

Consider the class C block 201.1.1.0 Give the design for 8 Subnets.

After Masking,

5	201.1.1.0	00000000	201.1.1.0
		⋮	
		00100000	201.1.1.32
		⋮	
		01000000	
10		⋮	201.1.1.64
		01100000	
		⋮	
		10000000	201.1.1.96
		⋮	
15		10100000	201.1.1.128
		⋮	
		11000000	201.1.1.160
		⋮	
		11100000	201.1.1.192
		⋮	
20		11110000	201.1.1.224

Netmask: 255.255.255.224

*25 Consider the class B block 140.60.0.0 Give the design for 256 Subnets. Give range of first 3 and last 3 address.

$$2^n = 256$$

$$n = 8$$

Default Class B Mask: 255.255.0.0 16 bit

Subnet Mask: 255.255.255.0 24 bit

host id. length = 8 bit

No of nodes addresses per Subnet. = $2^8 = 256$.

S_1 : 140.60.0.0 to 140.60.0.255

S_2 : 140.60.1.0 to 140.60.1.255

S_3 : 140.60.2.0 to 140.60.2.255

S_{254} : 140.60.253.0 to 140.60.253.255

S_{255} : 140.60.254.0 to 140.60.254.255

S_{256} : 140.60.255.0 to 140.60.255.255

Q. Consider the class B block 150.80.0.0. Give the design for 128 Subnets.

$$2^n = 128$$

$$n = 7$$

• Default Class B Mask : 255.255.0.0

• Subnet Mask : 255.255.254.0

• Host Id length : 9

No of Address per Subnet = $2^9 = 512$

S_1 : 150.80.0.0 to 150.80.1.255

S_2 : 150.80.2.0 to 150.80.3.255

S_3 : 150.80.4.0 to 150.80.5.255

S_{126} : 150.80.250.0 to 150.80.251.255

S_{127} : 150.80.252.0 to 150.80.253.255

S_{128} : 150.80.254.0 to 150.80.255.255

* Classless Addressing:

1. 2^n

2. First address of a block must be evenly divisible by no of IP addresses

If requirement < 256 only check rightmost octet.

→ If requirement < 65536 only check right 2 octet

If requirement > 65536 rightmost must be Zero.

→ If reqh is 0 then check 3rd octet is divisible by 4.
(if requirement multiple of 256)

ex if requirement = 1024, check 3rd Octet is divisible by 4

* Slash Notations:

x.y.z.t/n.

127
+ 27
154
+ 27
181

1. Prefix and Prefix length

2. Suffix and Suffix length



→ 150.30.16.81/27

Mask of 27 bits

150.30.16.81
255 255 255 224

150.30.16.64

Initial Address

150.30.16.64/27

32 device in network how?

↓
 $2^{32-27} = 2^5$
= 32

Q Find initial address for the following IP addresses

1) 130.40.16.91/26

2) 140.50.84.151/27

3) 120.50.32.0/23

1. $130.40.16.91/26$

$26 \Rightarrow 255.255.255.2$

bit in dot notation decimal

AND operation :

$130.40.16.91$

$255.255.255.2$

$130.40.16.64/26 \leftarrow \text{Initial Address}$

2. $140.50.84.151/27$

$27 \Rightarrow 255.255.255.3$

AND operation $\Rightarrow 140.50.84.128/27$

3. $120.50.32.0/23$

$23 \Rightarrow 255.255.254.0$

AND $\Rightarrow 120.50.32.0/23$

Q Find the initial & last address of the following IP address.

1. $120.30.16.151/6$

$6 \Rightarrow 252.0.0.0$

And Operation :

$120.30.16.151$

$252.0.0.0$

$120.30.16.151$

Host ID length : 26

$$1. 120.30.16.151/6 \rightarrow 120.30.16.128.$$

$$\text{Suffix: } \begin{array}{r} 00063 \\ \hline 120.30.16.191 \end{array}$$

$$2. 140.70.33.92/23.$$

23 bit Mask length.

(8) (8) (7) (0)

$$140.70.32.0.$$

$$\text{Suffix} = 4.$$

$$2^9 = 512. \rightarrow 511 = 1.255.255.255$$

$$140.70.32.0$$

$$0.0.0.1.255$$

$$140.70.33.255$$

$$3. 160.40.65.21/22.$$

$$22 \Rightarrow 255.255.252.0 \quad (\text{Mask})$$

$$160.40.65.21$$

$$\text{and } 255.255.252.0$$

$$160.40.64.0$$

$$(8) (8) (6) (0)$$

$$\text{Suffix} = 10$$

$$2^{10} =$$

* Clanten Subnetting

$$2^m = \text{No of Subnet}$$

$$\text{Let No of Subnet} = 4$$

$$2^m = 4$$

$$\boxed{m=2}$$

$$n = n + m \quad (\text{where } X.Y.Z.T/n)$$

$$\text{take } 111.32.16.64/26$$

$$\text{New } n = 26 + 2 = 28$$

$$\text{Suffix} = 32 - 28 = 4$$

$$\underline{2^4 = 16}$$

$$S1: 111.32.16.64/28 - 111.32.16.79/28$$

$$S2: 111.32.16.80/28 - 111.32.16.95/28$$

$$S3: 111.32.16.96/28 - 111.32.16.111/28$$

$$S4: 111.32.16.112/28 - 111.32.16.127/28$$

* Consider a class C block is Subnetted in 4 Subnets.

If one of the IP address is 195.168.1.79

Find its Subnet address.

$$\text{No of Subnet} = 4$$

$$\text{No of address per Subnet} = 2^{16-6} = 2^{10} = 1024$$

$$\text{Default Class C mask} = 255.255.255.0$$

$$\text{Subnet Mask} = 255.255.255.0$$

$$\text{Host Id length} = 6$$

$$\text{Initial Addr} = 195.168.1.64$$

b. Consider a Class B block is Subnetted into 128 Subnets
If one of the IP is 130.50.13.65 find Subnet address

No of Subnets = 128.

$$2^x = 128$$

$$x = 7$$

Default Mask = 255.255.0.0

Subnet Mask = 255.255.254.0

And on Subnet Mask & IP.

130.50.13.65

255.255.254.0

130.50.12.0

a. What is its direct broadcast address

⇒ ~~130.50.13.255~~ 130.50.12.0

0.0.0.1.255

130.50.13.255

a. 255.255.255.254 is the Subnet Mask, find no of Subnets. (Class C)

$$2^3 = 8$$

* Forwarding & Routing of IP Packets:

Direct Delivery & Indirect delivery.

Router is a machine which connects more than one network.

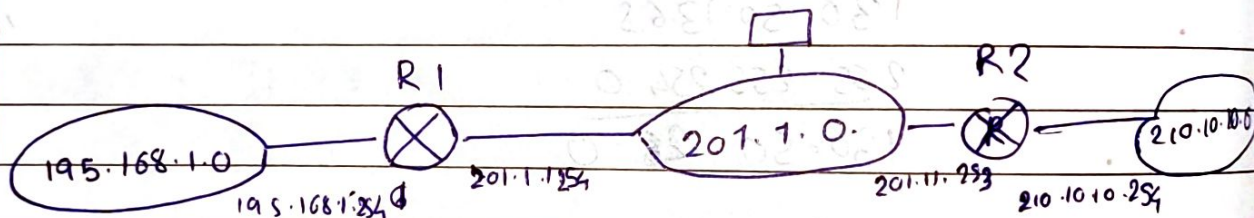
Router is also called as multi home computer, / gateway



router



host



src	dst	Delivery type,
195.168.1.1	195.168.1.10	direct delivery
195.168.1.1	201.10.10.100	Indirect delivery

Host forward to R1

R1 forward to R2

R2 deliver to recipient host.

How: forwarding is done?

With the help of routing table.

Routing Table for <195.168.1.1>

MASK	Network Address	Next Hop Address	Interface
255.255.255.0	195.168.1.0	—	195.168.1.1
255.255.255.0	201.1.1.0	195.168.1.254	195.168.1.1
255.255.255.0	210.10.10.0	195.168.1.254	195.168.1.1

R.T for <R1>

255.255.255.0	195.168.1.0	—	195.168.1.254
255.255.255.0	201.1.1.0	—	201.1.254
255.255.255.0	210.10.10.0	201.1.253	201.1.1.254

R.T for <R2>

255.255.255.0	210.10.10.0	—	210.10.10.254
255.255.255.0	201.1.1.0	—	201.1.1.253
255.255.255.0	195.168.1.0	201.1.1.254	201.1.1.253

1. SIP: 195.168.1.1 ← DIP: 195.168.1.10

195.168.1.10 $\xrightarrow{\text{Masking}}$ 195.168.1.0 (Same Network)

⇒ Next Hop Address: N.A.

(Direct Delivery)

Packet will be delivered by 195.168.1.1 to 195.168.1.10

As per RT of R1.

Next Hoping Address = 201.1.1.253.

- 2] Packet will be forwarded to R2 by R1
using interface (201.1.1.254).

As per RT of R2

Next Hoping Address = N.A.

- 3] Direct Delivery of packet by R2 to 210.10.10.100
using interface (210.10.10.254)

4. SIP: 210.10.10.100 DIP: 195.168.1.1

RT of <210.10.10.100>

Mask	Network Address	Next Hop Addr	Interface
255.255.255.0	210.10.10.0	—	210.10.10.100
255.255.255.0	201.1.1.0	200.10.10.253	210.10.10.100
255.255.255.0	195.168.1.0	200.10.10.254	210.10.10.100

195.168.1.1 —————> 195.168.1.0 (Other Network)
Mask (Indirect Delivery)

Next Hop Address: 210.10.10.254

- 1] Packet will be forwarded to R2 by 210.10.10.100

As per RT of R2

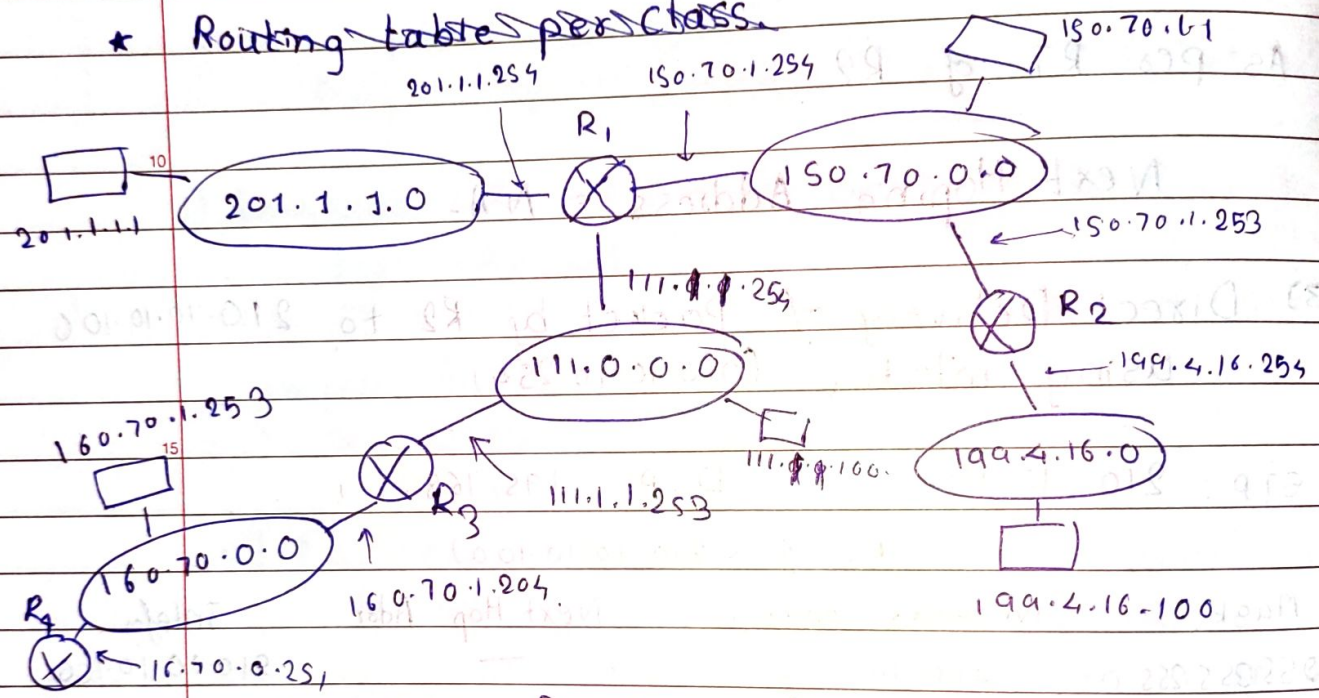
Next Hop Address: 201.1.1.254.

- 2] Packet will be forwarded by R2 to R1

As per RT of R1.
Next hopping Addr = NA.

Pa Direct Delivery of Packet by R1 to 195.168.1.1

* Routing table per class.



- * Default Router for R1 → R3
- * Default Router for R2 → R1
- * Default Router for R3 → R4

Routing table for <R1>

Mask	Network Address	Next Hoping Address	Interface
255.255.255.0	201.1.1.0	—	201.1.1.254
255.255.0.0	150.70.0.0	—	150.70.1.254
255.0.0.0	111.0.0.0	—	111.1.1.254
255.255.0.0	160.70.0.0	111.1.1.253	111.1.1.254
255.255.255.0	199.4.16.0	150.70.1.253	150.70.1.254
0.0.0.0	0.0.0.0	111.1.1.253	111.1.1.254

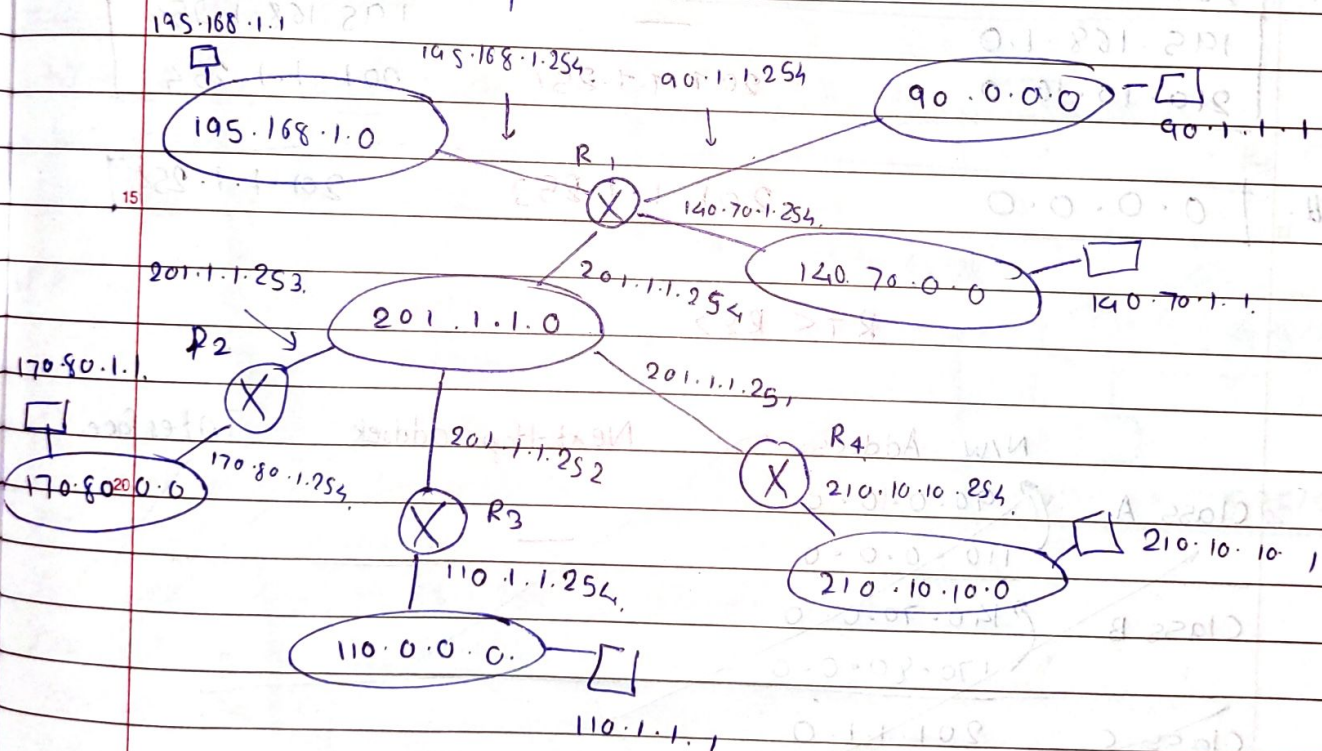
Routing Table < R2 >

Mask	Network Address	Next Hop Addr	Interface
	150.70.0.0	—	150.70.1.253
	199.4.16.0	—	199.4.16.254
	201.1.1.0	150.70.1.254	150.70.1.253
	160.70.0.0	150.70.1.254	150.70.1.253
0.0.0.0	0.0.0.0	150.70.1.254	15.70.1.253

Routing Table < R3 >

Network Addr	Next Hop Addr	Interface
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* Routing Table per Class:



* Routing Table for Router R1 → Algorithm.

1. Extract destination Ip
2. Determine its Class
3. Find its Network Address
4. Search appropriate routing table.

R	Default
R ₁	→ R ₂
R ₂	→ R ₃
R ₃	→ R ₄
R ₄	→

8

5

RT < R1 > For class A

Class A

N/w address

Next-Top address

Interface

90.0.0.0

110.0.0.0

201.1.1.252

90.1.1.254

201.1.1.254

Class B

140.70.0.0

170.80.0.0

201.1.1.253

140.70.1.254

201.1.1.254

Class C

201.1.1.0

195.168.1.0

210.10.10.0

201.1.1.251

201.1.1.254

195.168.1.254

201.1.1.254

default

0.0.0.0

201.1.1.253

201.1.1.254

RT < R2 >

N/W Address

Next-Top address

Interface

Class A

90.0.0.0

110.0.0.0

Class B

140.70.0.0

170.80.0.0

Class C

201.1.1.0

195.1

RT < R2 >

Class A

110.0.0.0

90.0.0.0

201.1.1.254

201.1.1.254

201.1.1.253

201.1.1.254

Class B

170.80.0.0

140.70.0.0

201.1.1.254

170.80.1.254

201.1.1.253

Class C

201.1.1.0

195.168.1.0

210.10.10.0

201.1.1.254

201.1.1.251

201.1.1.253

201.1.1.253

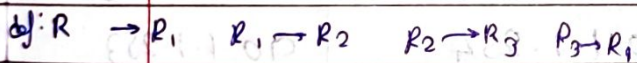
201.1.1.253

Default

0.0.0.0

201.1.1.253

201.1.1.253



$\langle 195.168.1.1/24 \rangle RT$

In	network	Next Hop	Interface
Address	Address		
128	130.40.4.32	195.168.1.254	195.168.1.1
126	120.80.4.64	195.168.1.254	195.168.1.1
125	111.4.16.128	195.168.1.254	195.168.1.1
124	195.168.1.0	—	195.168.1.1
116	140.80.0.0	195.168.1.254	195.168.1.1
18	40.0.0.0	195.168.1.254	195.168.1.1
10 (default)	0.0.0.0	195.168.1.254	195.168.1.1

<R1> RT

in	Network Address	Next Hop Address	Interface
128	130.40.4.32	90.1.1.251	90.1.1.251
126	120.80.4.64	90.1.1.253	90.1.1.253
125	111.4.16.128	90.1.1.254	90.1.1.254
124	195.168.1.0	—	195.168.1.284
116	140.80.0.0	—	140.80.1.254
18	90.0.0.0	—	90.1.1.254
10	0.0.0.0	90.1.1.253	90.1.1.254

<R3> RT

in	Network Address	Next Hop Addr	Interface
128	130.40.4.32	90.1.1.251	90.1.1.253
126	120.80.4.64	—	120.80.4.108
125	111.4.16.128	90.1.1.251	90.1.1.253
124	195.168.1.0	90.1.1.254	90.1.1.253
116	140.80.0.0	90.1.1.254	90.1.1.253
18	90.0.0.0	—	90.1.1.253
10	0.0.0.0	90.1.1.251	90.1.1.253

<R3> RT

in	Network Address	Next Hop Addr	Interface
128	130.40.4.32	111.4.16.253	111.4.16.254
126	120.80.4.64	90.1.1.253	90.1.1.251
125	111.4.16.128	—	111.4.16.254
124	195.168.1.0	90.1.1.254	90.1.1.251
116	140.80.0.0	90.1.1.254	90.1.1.251
18	90.0.0.0	—	90.1.1.251
10	0.0.0.0	111.4.16.253	